MRSIPrep report: SynthMRSI-Project · sub-01 ses-01

Inputs

BIDS directory: `/data`

Output directory: `/out/mrsiprep`

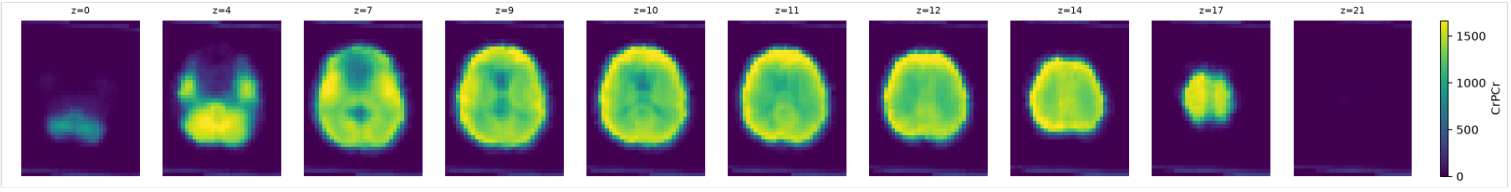
Parcellation mode: `synthseg`

Tissue backend: `synthseg-fast`

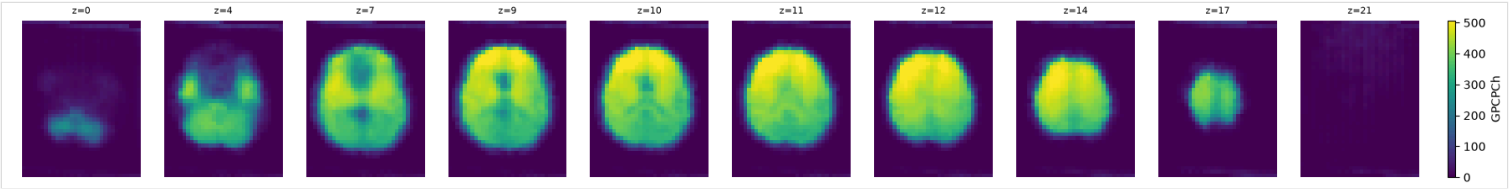
MRSI Raw

Raw metabolite maps (pre-pipeline)

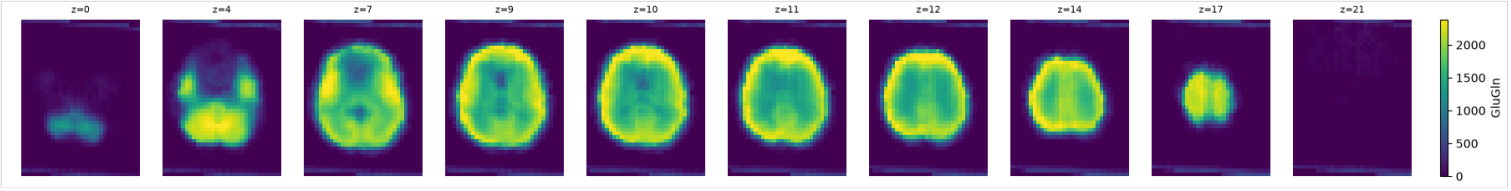
Metabolite: CrPCr



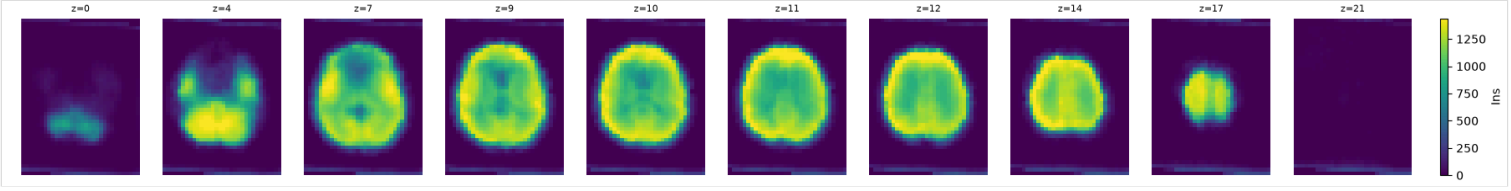
Metabolite: GPCPCh



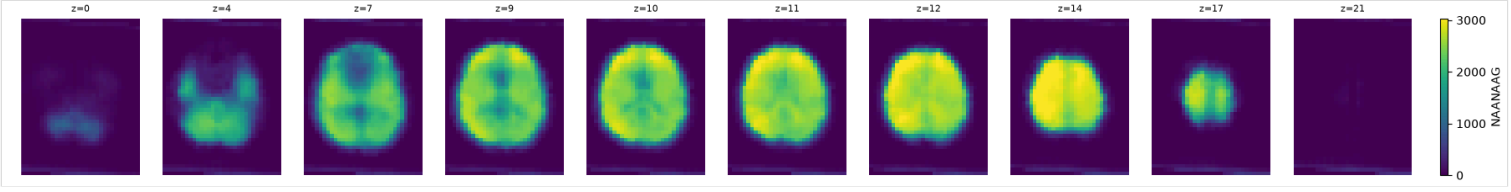
Metabolite: GluGln



Metabolite: Ins



Metabolite: NAANAAG



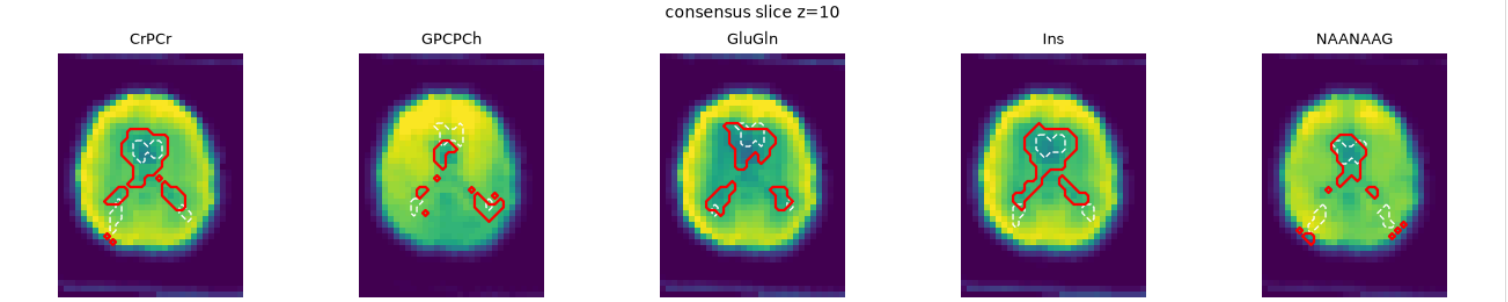
MRSI Raw QC

metabolite	n_total_voxels	mean_snr	median_snr	mean_linewidth	median_linewidth	mean_crlb	median_crlb
NAANAAG	20738	14.135942	15.0	0.038052	0.036	4.933007	4.0
GPCPCh	20738	14.228251	15.0	0.037878	0.036	6.853189	6.0
CrPCr	20738	14.257129	15.0	0.037799	0.036	6.431937	5.0
GluGln	20738	14.345459	15.0	0.037970	0.036	11.069957	11.0

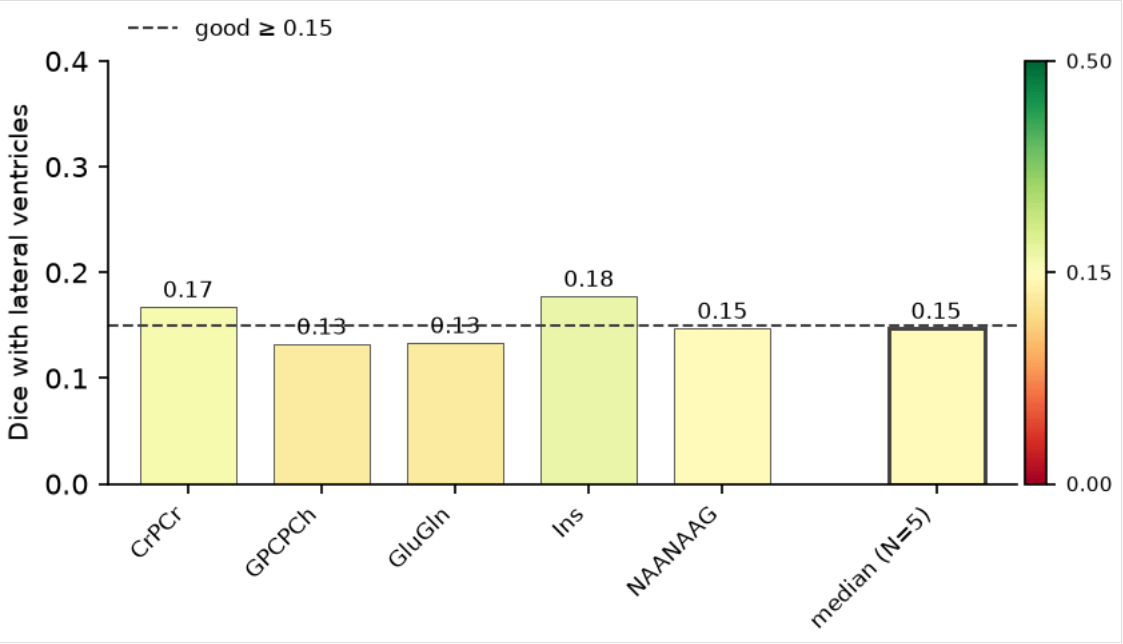
metabolite	n_total_voxels	mean_snr	median_snr	mean_linewidth	median_linewidth	mean_crlb	median_crlb
Ins	20738	14.465190	15.0	0.037318	0.033	8.926435	8.0

Ventricle visibility

detected ventricle MNI prior native MRSI space, pre-coregistration · all metabolites at z=10



Dice with the subject's lateral ventricles · median 0.15 (SynthSeg, moved to MRSI→T1w transform; green from 0.15)

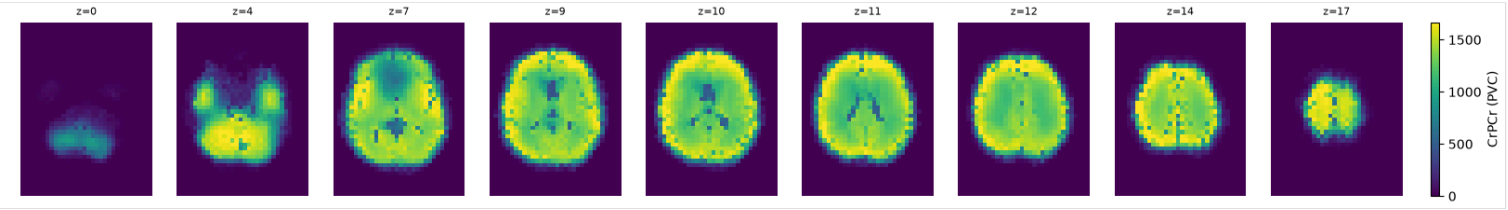


MRSI PVC

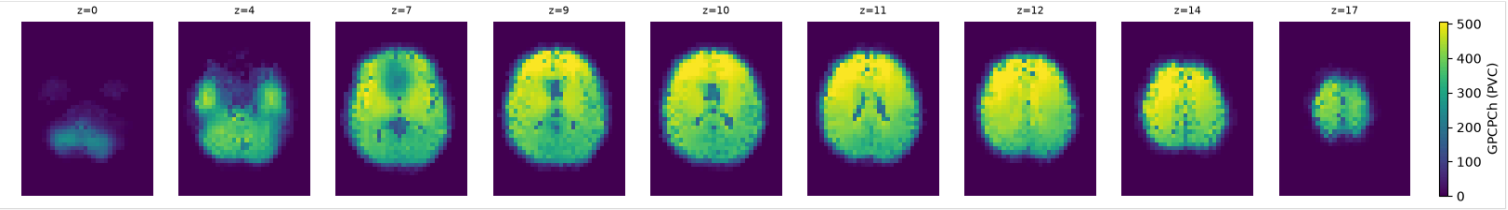
Partial-volume correction

Partial-volume-corrected metabolite maps, on the native MRSI grid, at the same slices as the MRSI Raw tab so the two can be compared directly.

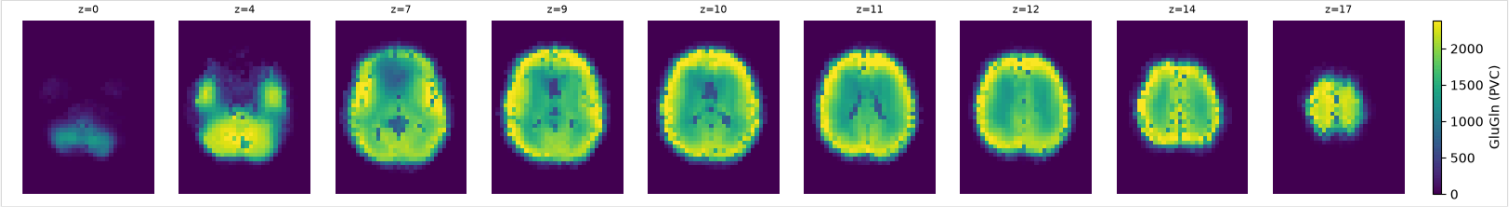
Metabolite: CrPCr



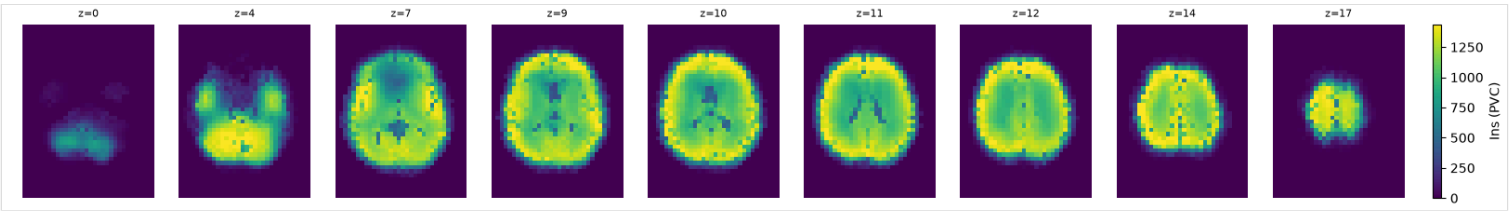
Metabolite: GPCPCh



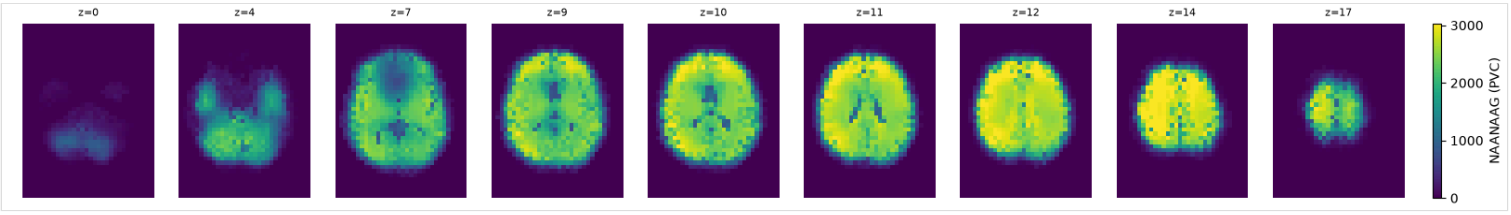
Metabolite: GluGln



Metabolite: Ins



Metabolite: NAANAAG



Spike Filter

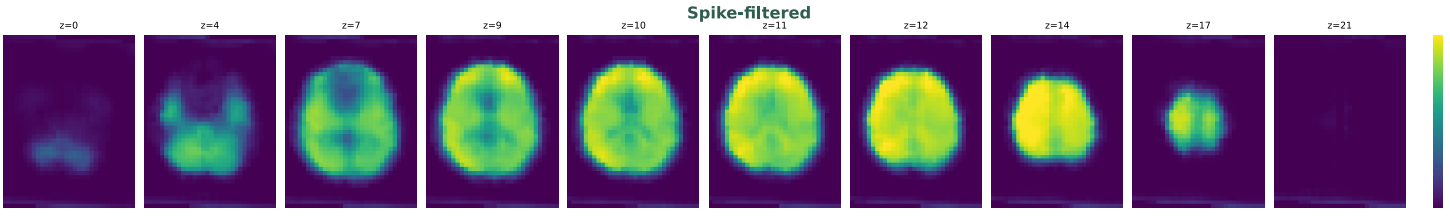
Filtering summary

Metabolite	Spike voxels detected
NAANAAG	0
GPCPCh	0
CrPCr	7
GluGln	25
Ins	20

Metabolite: NAANAAG

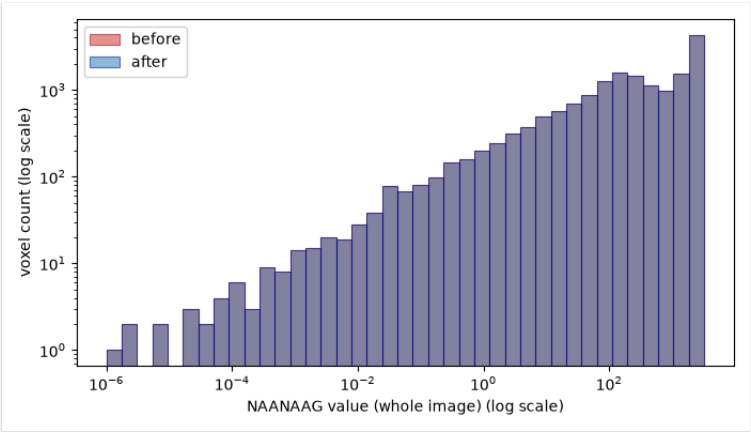
Spike voxels detected: 0

Hover over the figure to flip between before and after spike filtering.

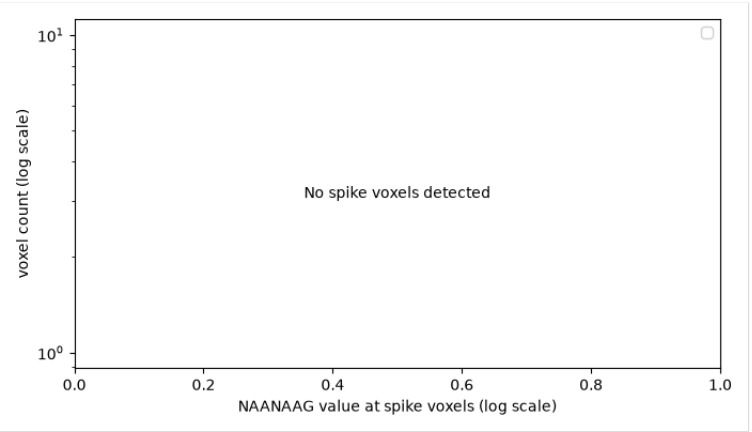


Get figure file: figures/sub-01_ses-01_step-mrsi-preproc_met-NAANAAG_spikefilt.svg

Whole image



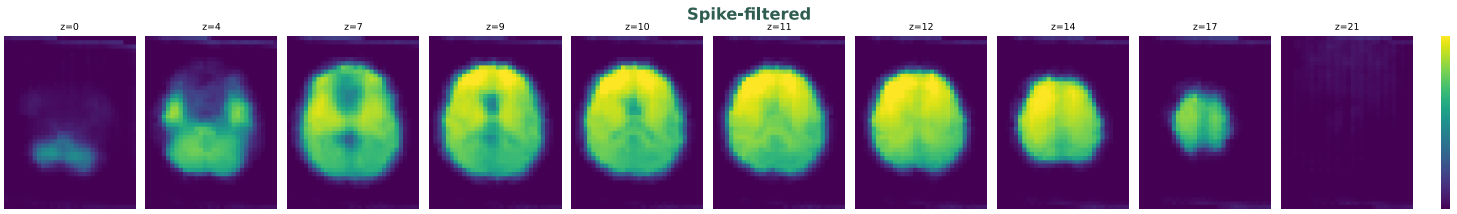
Spike voxels only



Metabolite: GPCPCh

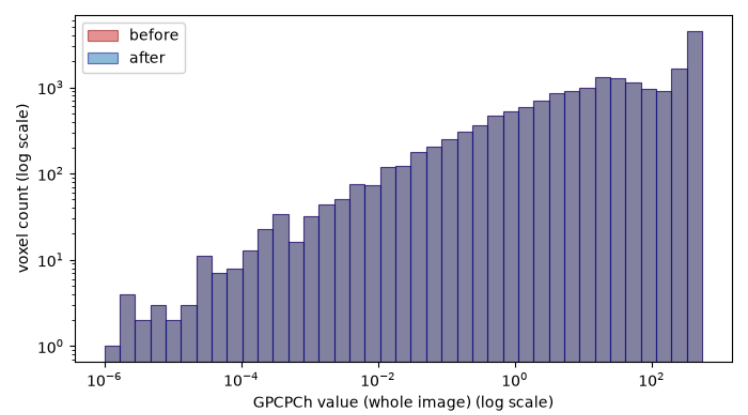
Spike voxels detected: 0

Hover over the figure to flip between before and after spike filtering.

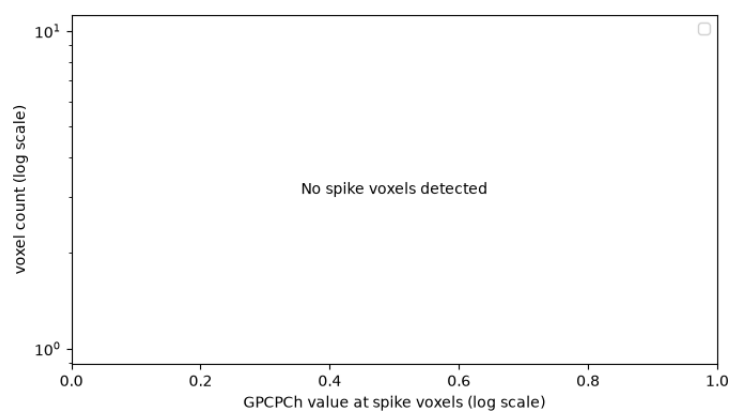


Get figure file: figures/sub-01_ses-01_step-mrsi-preproc_met-GPCPCh_spikefilt.svg

Whole image



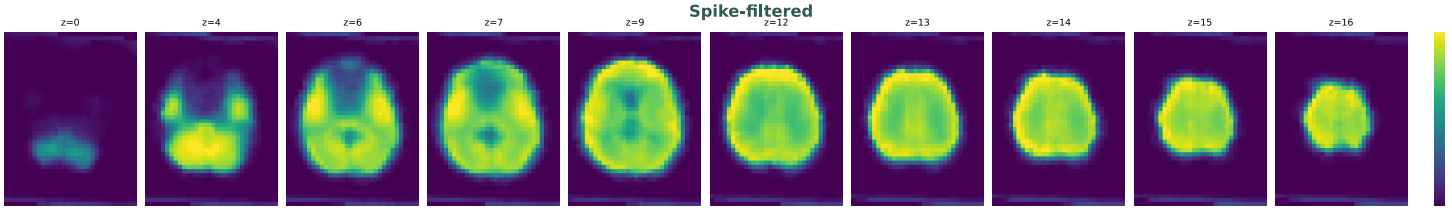
Spike voxels only



Metabolite: CrPCr

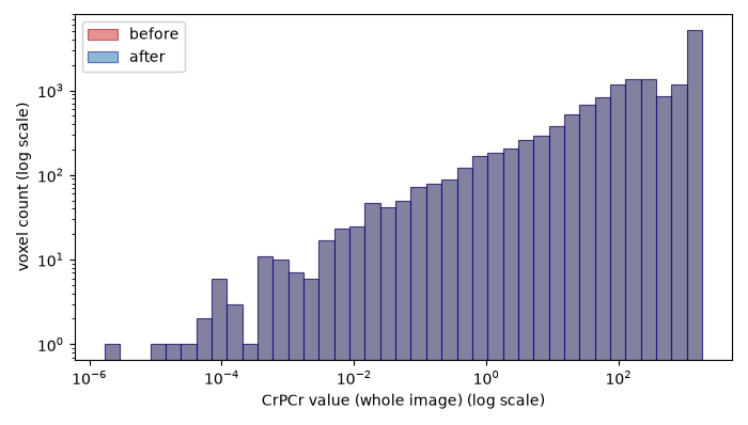
Spike voxels detected: 7

Hover over the figure to flip between before and after spike filtering.

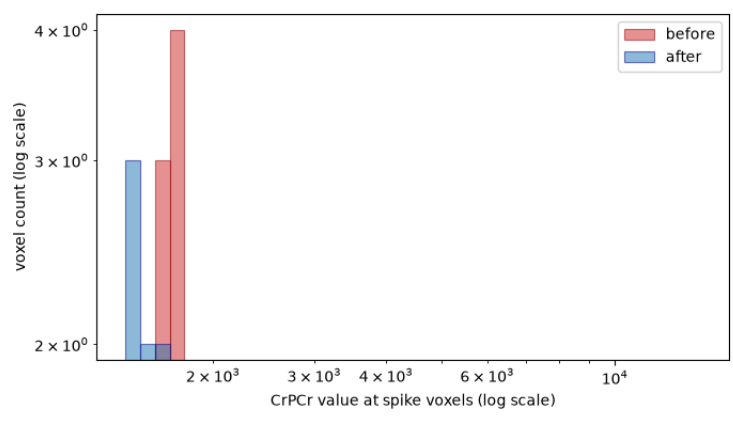


Get figure file: figures/sub-01_ses-01_step-mrsi-preproc_met-CrPCr_spikefilt.svg

Whole image



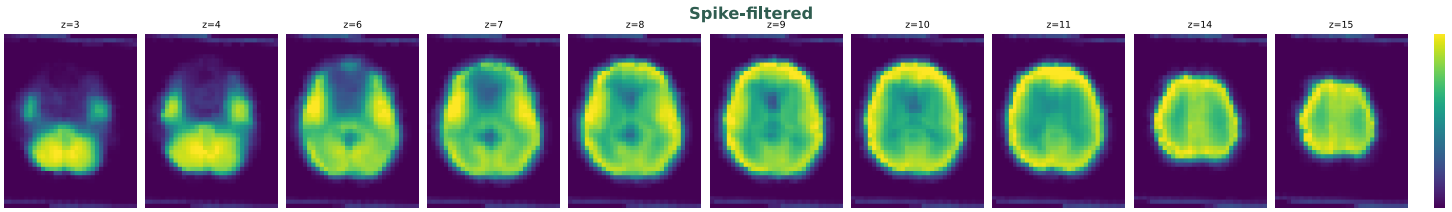
Spike voxels only



Metabolite: GluGln

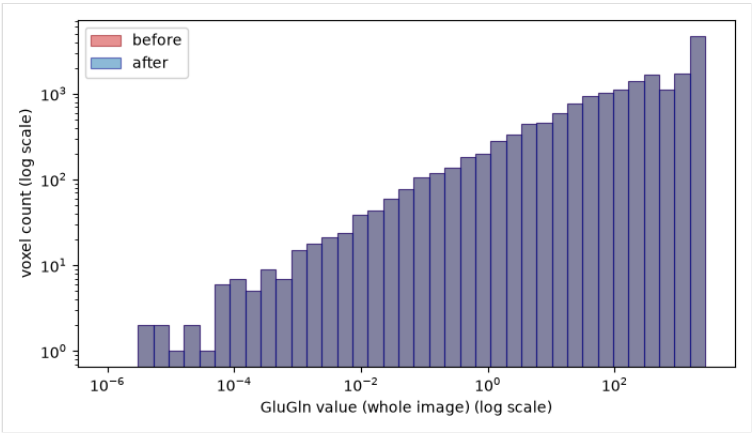
Spike voxels detected: 25

Hover over the figure to flip between before and after spike filtering.

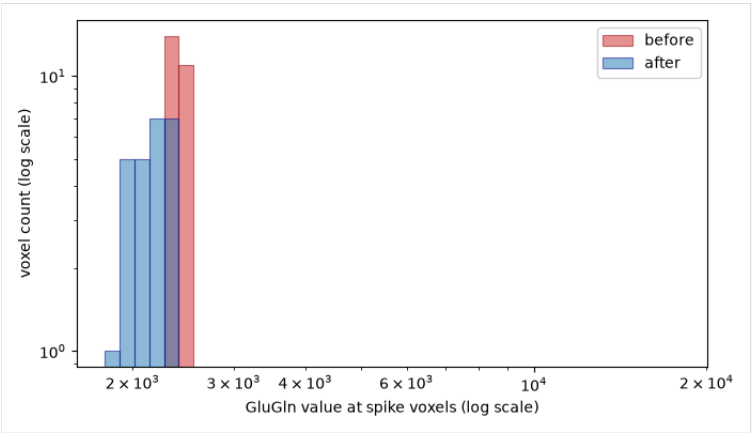


Get figure file: figures/sub-01_ses-01_step-mrsi-preproc_met-GluGln_spikefilt.svg

Whole image



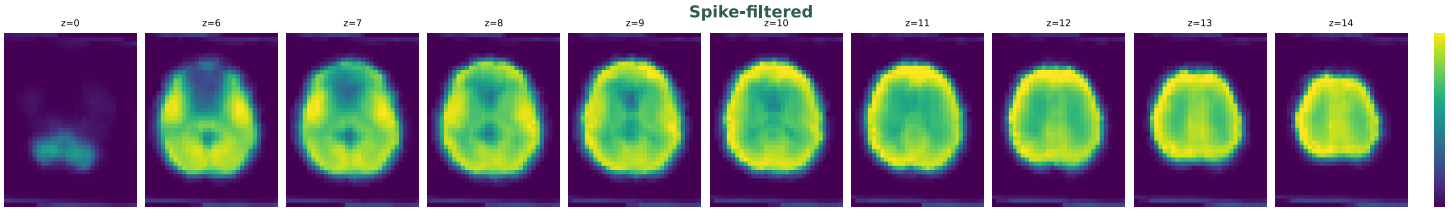
Spike voxels only



Metabolite: Ins

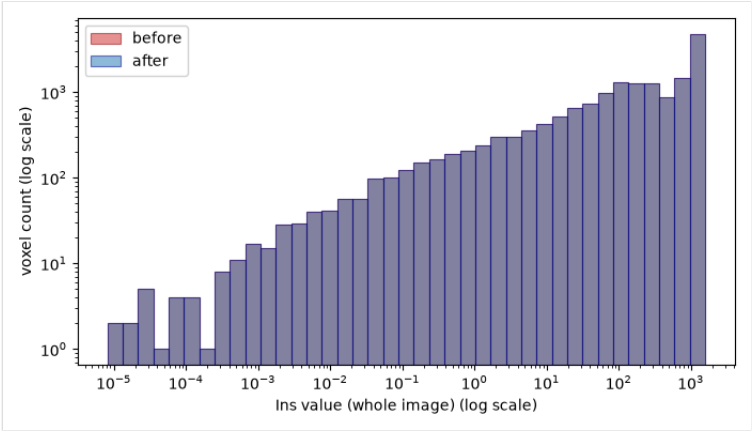
Spike voxels detected: 20

Hover over the figure to flip between before and after spike filtering.

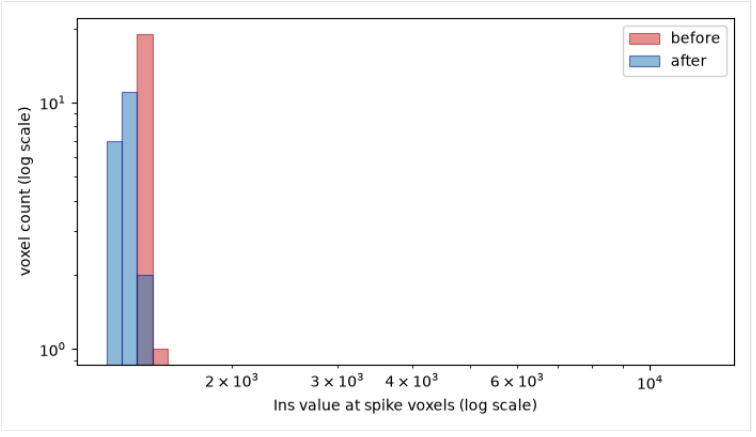


Get figure file: figures/sub-01_ses-01_step-mrsi-preproc_met-Ins_spikefilt.svg

Whole image

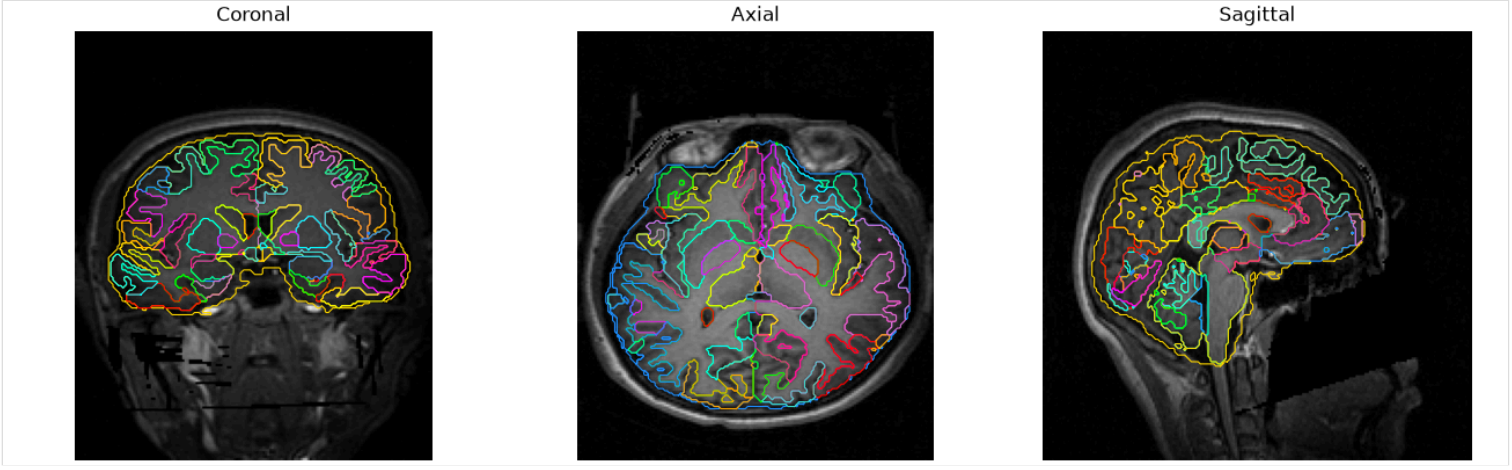


Spike voxels only



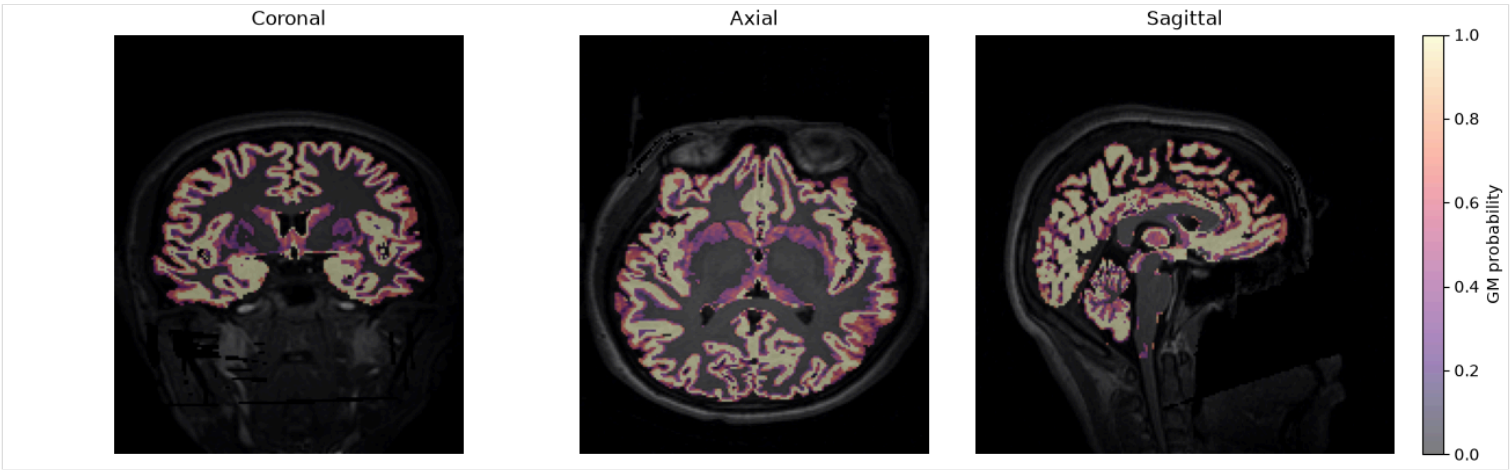
Anatomical

Tissue label outlines

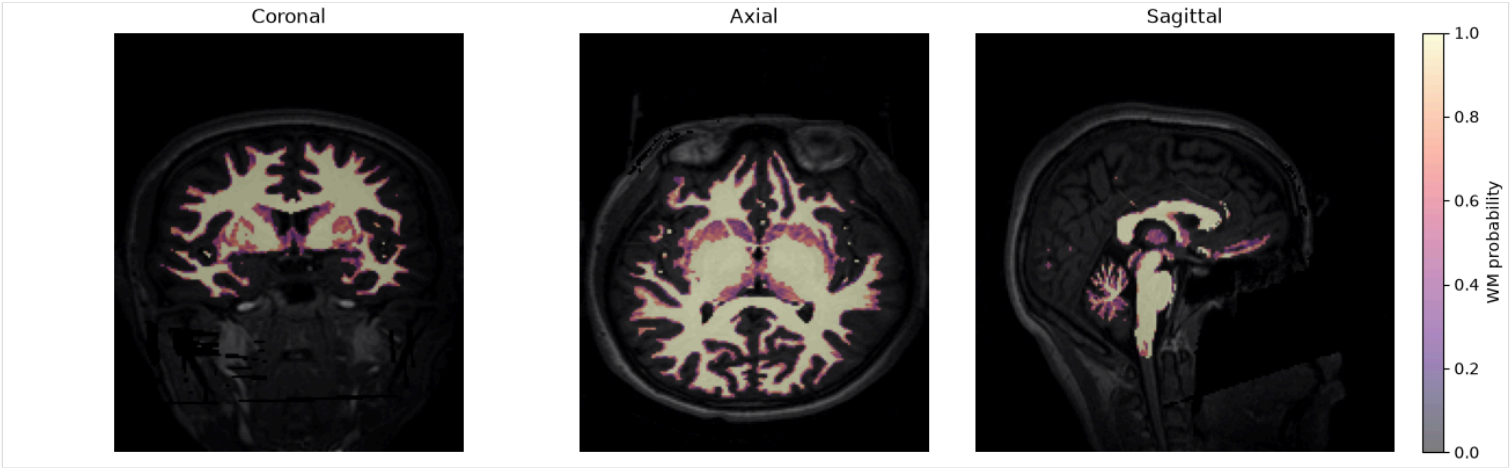


Tissue probability maps

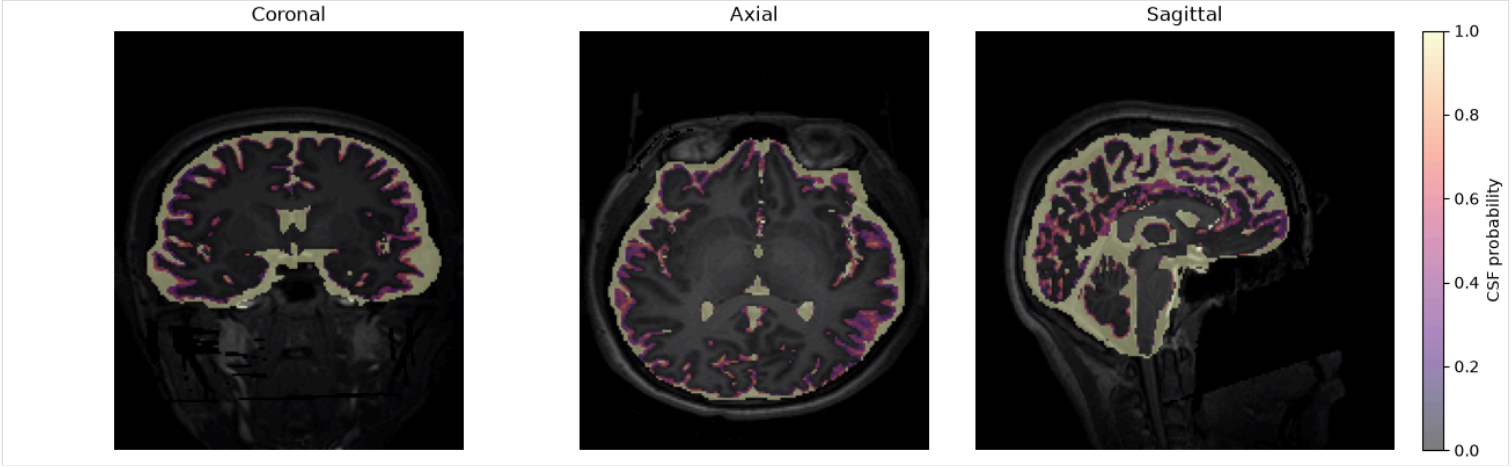
GM



WM



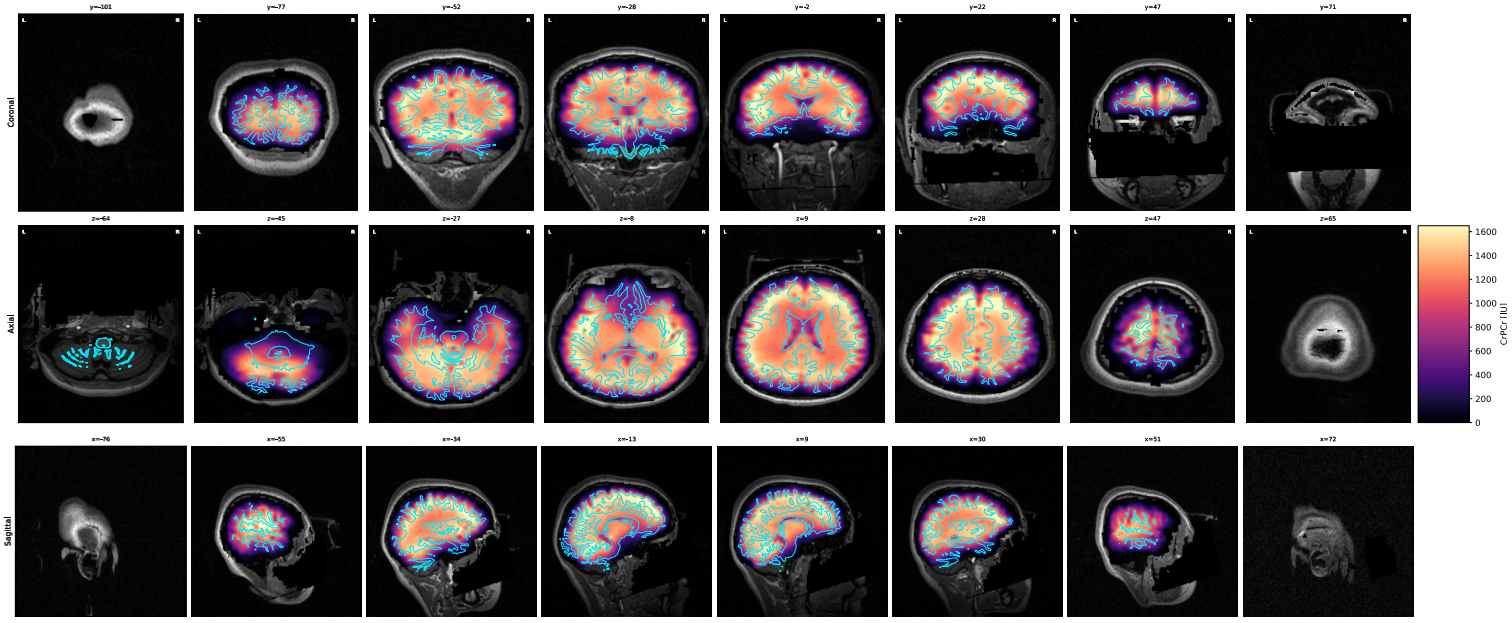
CSF



T1-space alignment

T1w-space alignment

Reference metabolite map (`CrPCr`) overlaid on raw T1w, with the white-matter boundary in green. Hover over the figure to flip between the two.



Get figure file: figures/sub-01_ses-01_step-registration_space-T1w.svg

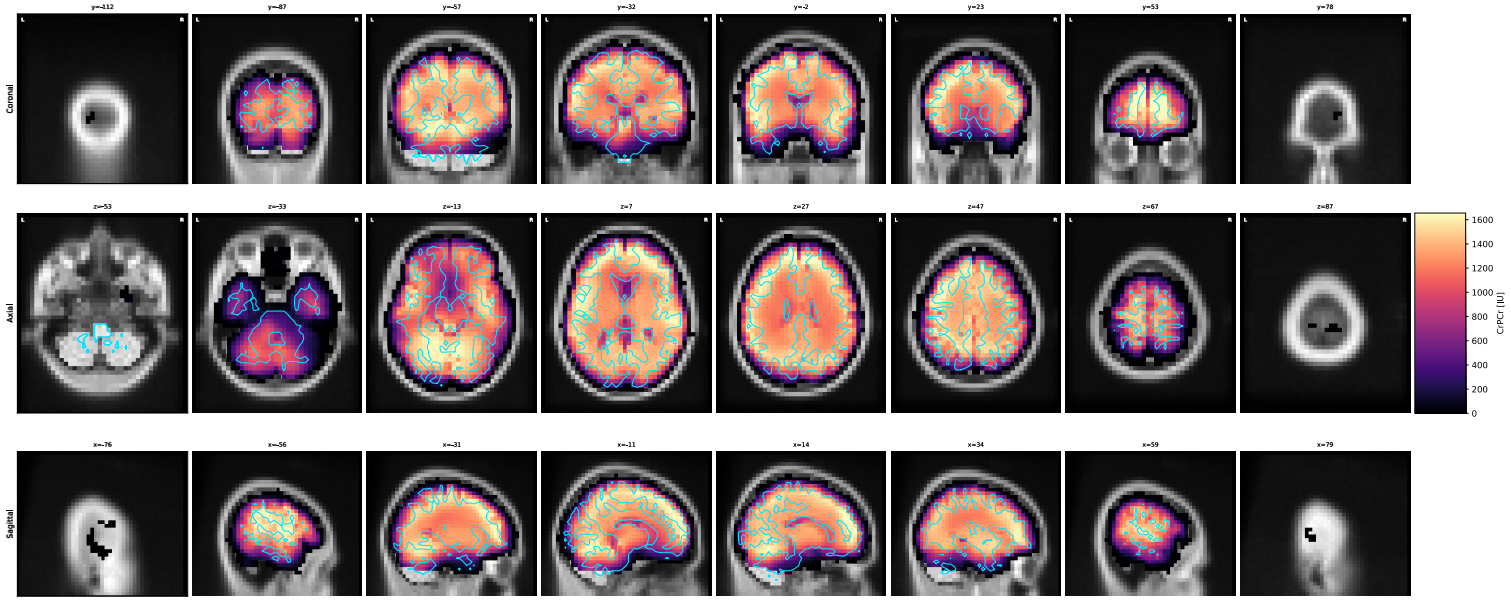
Signal-weighted leakage

Not computed: T1w-space leakage requires the full T1w-space resampled metabolite maps (`--output-mrsi-t1w`), which weren't requested for this run. The alignment image above uses a separate, lightweight report-only resampling and doesn't imply this data is available.

Template-space alignment

Template-space alignment

Reference metabolite map (`CrPCr`) overlaid on the full-head `MNI152NLin2009cAsym` template at 5 mm, with the white-matter boundary in green. Hover over the figure to flip between the two.



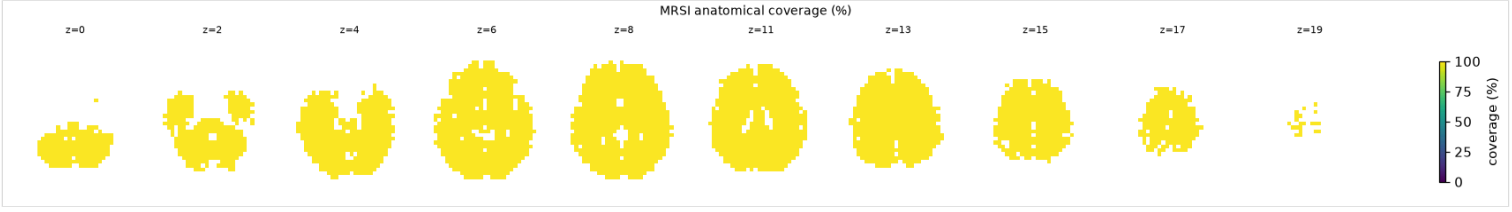
Get figure file: figures/sub-01_ses-01_step-registration_space-MNI.svg

Signal-weighted leakage

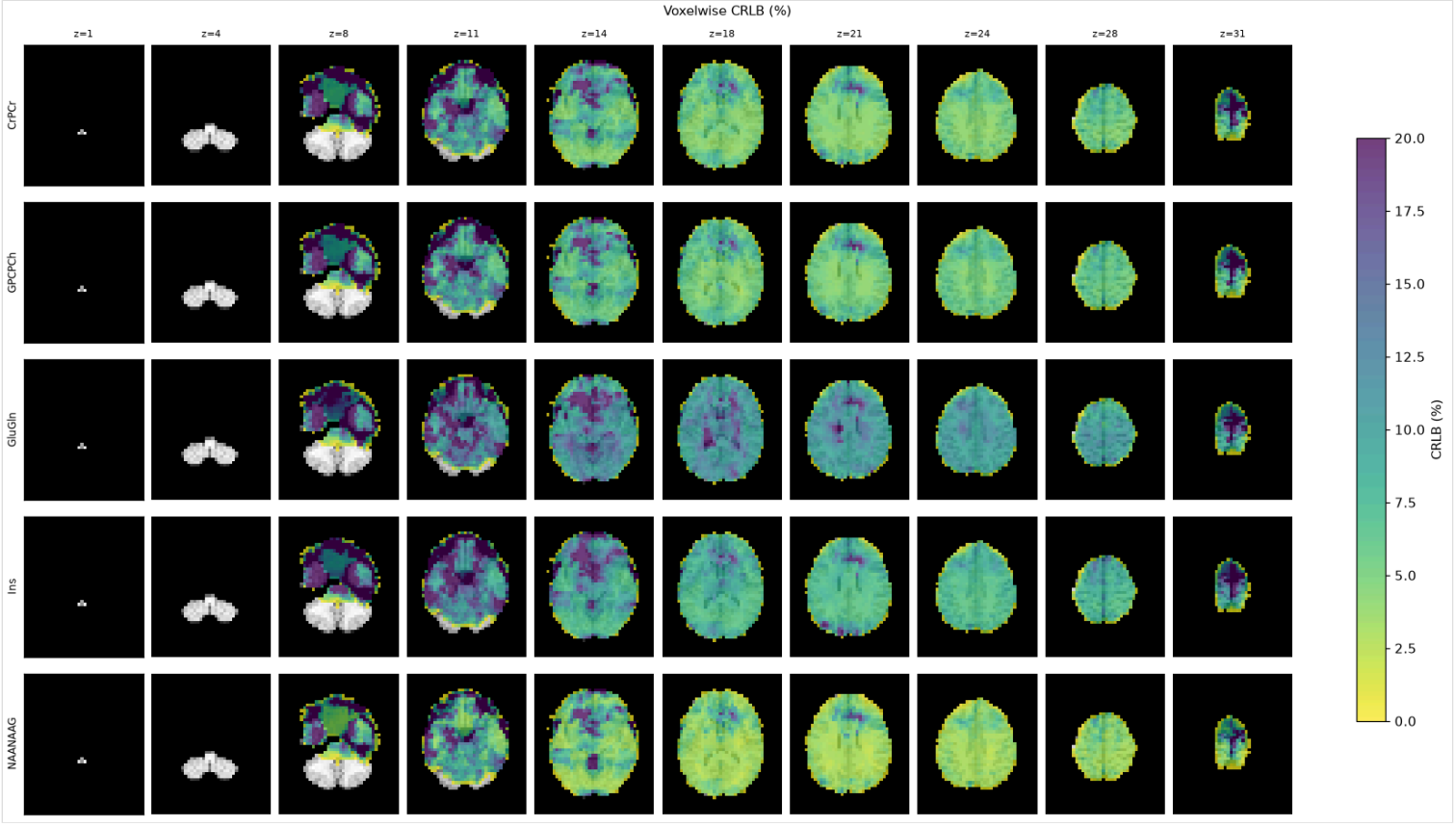
metabolite	n_quality_voxels	total_signal_mass	leakage_signal_mass	leakage_fraction	leakage_percent
CrPCr	73723	15157284.000	77581.797	0.005	0.512
GPCPCh	73883	4231608.500	25804.670	0.006	0.610
GluGln	73608	20610924.000	127430.555	0.006	0.618
Ins	73366	12834889.000	76472.367	0.006	0.596
NAANAAG	74263	23854156.000	131025.789	0.005	0.549

Coverage

MRSI anatomical coverage



Parcelwise CRLB quality (green reliable / red unreliable)



`qc_valid_fraction` is the fraction of a parcel's MRSI voxels that passed every per-voxel quality threshold in force for this run (CRLB ≤ 20.0%, SNR ≥ 4.0, linewidth ≤ 0.1), averaged over metabolites. Rows are ordered worst first.

parcel_id	parcel_name	hemisphere	mean_crlb	qc_valid_fraction
1006	ctx-lh-entorhinal	L	112.650	0.150
46	Right-Cerebellum-White-Matter	R	22.233	0.270
18	Left-Amygdala	L	21.673	0.273
1032	ctx-lh-frontalpole	L	161.900	0.289
2006	ctx-rh-entorhinal	R	41.380	0.300
2032	ctx-rh-frontalpole	R	317.525	0.311
7	Left-Cerebellum-White-Matter	L	13.242	0.329
1033	ctx-lh-temporalpole	L	101.237	0.388
47	Right-Cerebellum-Cortex	R	29.372	0.405
1009	ctx-lh-inferiortemporal	L	37.039	0.422
8	Left-Cerebellum-Cortex	L	16.590	0.437
2033	ctx-rh-temporalpole	R	122.071	0.486
16	Brain-Stem	NaN	16.152	0.551
2019	ctx-rh-parsorbitalis	R	41.533	0.560
26	Left-Accumbens-area	L	94.300	0.567
1011	ctx-lh-lateraloccipital	L	8.468	0.578
1027	ctx-lh-rostralmiddlefrontal	L	8.896	0.639
2027	ctx-rh-rostralmiddlefrontal	R	24.775	0.690
2009	ctx-rh-inferiortemporal	R	42.380	0.705
1015	ctx-lh-middletemporal	L	21.457	0.705
2011	ctx-rh-lateraloccipital	R	7.613	0.731
58	Right-Accumbens-area	R	17.200	0.733
1007	ctx-lh-fusiform	L	33.067	0.756
2012	ctx-rh-lateralorbitofrontal	R	27.377	0.765
2002	ctx-rh-caudalanteriorcingulate	R	14.181	0.790
1012	ctx-lh-lateralorbitofrontal	L	23.344	0.808
1019	ctx-lh-parsorbitalis	L	11.787	0.812
1031	ctx-lh-supramarginal	L	6.535	0.816
17	Left-Hippocampus	L	12.923	0.838

parcel_id	parcel_name	hemisphere	mean_crlb	qc_valid_fraction
1022	ctx-lh-postcentral	L	6.697	0.857
1028	ctx-lh-superiorfrontal	L	7.874	0.858
60	Right-VentralDC	R	10.807	0.869
1016	ctx-lh-parahippocampal	L	12.118	0.871
1003	ctx-lh-caudalmiddlefrontal	L	6.991	0.880
1002	ctx-lh-caudalanteriorcingulate	L	11.558	0.883
2014	ctx-rh-medialorbitofrontal	R	15.297	0.892
1029	ctx-lh-superiorparietal	L	6.991	0.896
11	Left-Caudate	L	9.455	0.897
54	Right-Amygdala	R	10.483	0.900
1014	ctx-lh-medialorbitofrontal	L	19.058	0.902
13	Left-Pallidum	L	10.956	0.911
2007	ctx-rh-fusiform	R	15.569	0.913
1008	ctx-lh-inferiorparietal	L	7.046	0.920
2029	ctx-rh-superiorparietal	R	6.856	0.920
53	Right-Hippocampus	R	10.333	0.924
1030	ctx-lh-superiortemporal	L	7.635	0.926
12	Left-Putamen	L	8.739	0.928
2026	ctx-rh-rostralanteriorcingulate	R	11.872	0.928
2016	ctx-rh-parahippocampal	R	10.188	0.929
2028	ctx-rh-superiorfrontal	R	7.397	0.930
2003	ctx-rh-caudalmiddlefrontal	R	7.837	0.931
2015	ctx-rh-middletemporal	R	14.442	0.933
2031	ctx-rh-supramarginal	R	6.279	0.934
50	Right-Caudate	R	9.443	0.935
28	Left-VentralDC	L	12.368	0.936
1024	ctx-lh-precentral	L	7.056	0.943
2035	ctx-rh-insula	R	7.328	0.947
2	Left-Cerebral-White-Matter	L	9.903	0.953
2024	ctx-rh-precentral	R	6.865	0.955
1026	ctx-lh-rostralanteriorcingulate	L	10.450	0.957
1035	ctx-lh-insula	L	7.397	0.965
2017	ctx-rh-paracentral	R	7.300	0.967
2022	ctx-rh-postcentral	R	6.707	0.967
2030	ctx-rh-superiortemporal	R	6.940	0.972
2020	ctx-rh-parstriangularis	R	9.964	0.973
1017	ctx-lh-paracentral	L	7.351	0.973
41	Right-Cerebral-White-Matter	R	7.787	0.977
52	Right-Pallidum	R	9.667	0.978
1013	ctx-lh-lingual	L	6.558	0.984
51	Right-Putamen	R	7.344	0.989
1018	ctx-lh-parsopercularis	L	7.663	0.990
10	Left-Thalamus	L	7.531	0.993
1020	ctx-lh-parstriangularis	L	8.022	0.994
2013	ctx-rh-lingual	R	6.992	0.996
49	Right-Thalamus	R	6.957	0.997
1025	ctx-lh-precuneus	L	5.997	0.997
1001	ctx-lh-bankssts	L	6.062	1.000
1023	ctx-lh-posteriorcingulate	L	6.032	1.000
1005	ctx-lh-cuneus	L	6.712	1.000
1010	ctx-lh-isthmuscingulate	L	5.926	1.000
2008	ctx-rh-inferiorparietal	R	6.301	1.000

parcel_id	parcel_name	hemisphere	mean_crlb	qc_valid_fraction
2005	ctx-rh-cuneus	R	6.538	1.000
2001	ctx-rh-bankssts	R	6.214	1.000
1034	ctx-lh-transversetemporal	L	5.911	1.000
1021	ctx-lh-pericalcarine	L	7.150	1.000
2021	ctx-rh-pericalcarine	R	6.200	1.000
2018	ctx-rh-parsopercularis	R	7.855	1.000
2010	ctx-rh-isthmuscingulate	R	5.882	1.000
2023	ctx-rh-posteriorcingulate	R	6.000	1.000
2025	ctx-rh-precuneus	R	5.816	1.000
2034	ctx-rh-transversetemporal	R	5.717	1.000

Parcellation

Not available for this configuration.

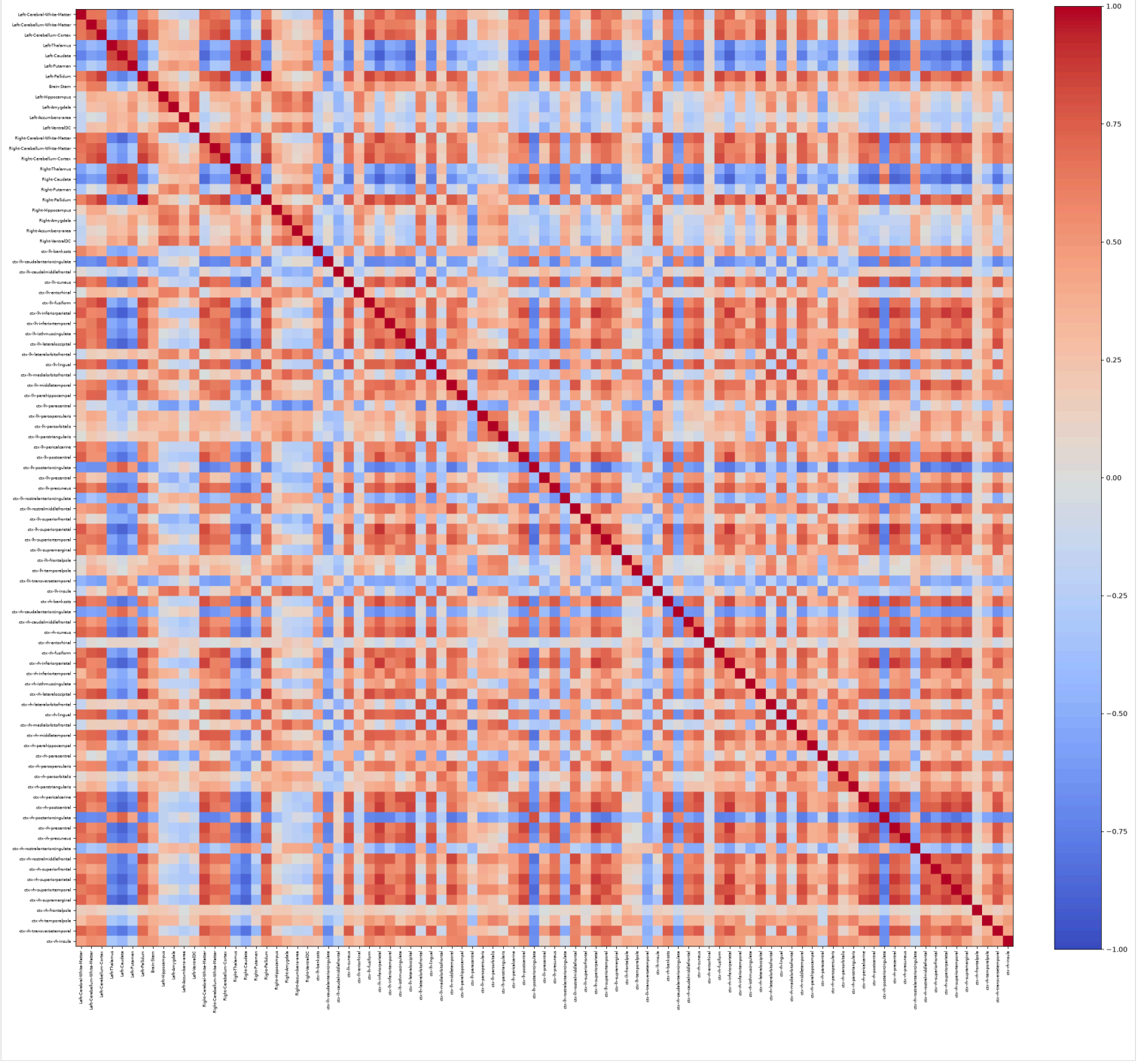
Regional metabolites: synthseg

subject	session	atlas	scale	parcel_id	parcel_name	hemisphere	metabolite	mean	median	std	weighted_mean
sub-01	ses-01	synthseg	NaN	2	Left-Cerebral-White-Matter	L	NAANAAG	2396.673584	2528.545898	633.721252	2462.376953
sub-01	ses-01	synthseg	NaN	2	Left-Cerebral-White-Matter	L	GPCPCh	403.141846	417.577026	90.989296	405.740936
sub-01	ses-01	synthseg	NaN	2	Left-Cerebral-White-Matter	L	CrPCr	1354.009033	1378.217407	268.945099	1370.252930
sub-01	ses-01	synthseg	NaN	2	Left-Cerebral-White-Matter	L	GluGln	1819.495483	1836.212646	418.186066	1841.236938
sub-01	ses-01	synthseg	NaN	2	Left-Cerebral-White-Matter	L	Ins	1155.218506	1178.524780	232.018356	1167.601807
sub-01	ses-01	synthseg	NaN	7	Left-Cerebellum-White-Matter	L	NAANAAG	1601.608154	1682.506104	352.608459	1600.009277
sub-01	ses-01	synthseg	NaN	7	Left-Cerebellum-White-Matter	L	GPCPCh	327.797089	336.766632	40.411880	327.713379
sub-01	ses-01	synthseg	NaN	7	Left-Cerebellum-White-Matter	L	CrPCr	1312.124390	1342.622314	230.455032	1310.596924
sub-01	ses-01	synthseg	NaN	7	Left-Cerebellum-White-Matter	L	GluGln	1794.603149	1851.894897	346.419037	1789.484741
sub-01	ses-01	synthseg	NaN	7	Left-Cerebellum-White-Matter	L	Ins	1147.921021	1193.803467	190.169586	1146.533203
sub-01	ses-01	synthseg	NaN	8	Left-Cerebellum-Cortex	L	NAANAAG	1766.620361	1872.360352	459.101074	1805.566650
sub-01	ses-01	synthseg	NaN	8	Left-Cerebellum-Cortex	L	GPCPCh	335.426758	359.033813	68.881439	340.346588
sub-01	ses-01	synthseg	NaN	8	Left-Cerebellum-Cortex	L	CrPCr	1403.692261	1506.588135	324.494202	1433.088501
sub-01	ses-01	synthseg	NaN	8	Left-Cerebellum-Cortex	L	GluGln	1939.174072	2087.803955	463.386749	1982.745483
sub-01	ses-01	synthseg	NaN	8	Left-Cerebellum-Cortex	L	Ins	1243.190186	1309.930786	242.067902	1260.809570

subject	session	atlas	scale	parcel_id	parcel_name	hemisphere	metabolite	mean	median	std	weighted_mean
sub-01	ses-01	synthseg	NaN	10	Left-Thalamus	L	NAANAAG	2057.937256	2047.638550	210.116440	2059.058838
sub-01	ses-01	synthseg	NaN	10	Left-Thalamus	L	GPCPCh	385.179779	388.949036	43.590355	384.806458
sub-01	ses-01	synthseg	NaN	10	Left-Thalamus	L	CrPCr	1239.272705	1259.984741	96.799187	1237.261353
sub-01	ses-01	synthseg	NaN	10	Left-Thalamus	L	GluGln	1544.851685	1563.124878	112.994614	1542.379883
sub-01	ses-01	synthseg	NaN	10	Left-Thalamus	L	Ins	1005.802124	1018.390564	66.916145	1004.504028
sub-01	ses-01	synthseg	NaN	11	Left-Caudate	L	NAANAAG	2010.669067	2090.767090	328.459320	2060.323242
sub-01	ses-01	synthseg	NaN	11	Left-Caudate	L	GPCPCh	394.171906	404.902252	42.871990	399.732483
sub-01	ses-01	synthseg	NaN	11	Left-Caudate	L	CrPCr	1076.567383	1079.396851	113.229401	1086.255493
sub-01	ses-01	synthseg	NaN	11	Left-Caudate	L	GluGln	1236.602173	1232.643799	150.012558	1251.919922
sub-01	ses-01	synthseg	NaN	11	Left-Caudate	L	Ins	864.418335	864.684570	93.727417	873.180603
sub-01	ses-01	synthseg	NaN	12	Left-Putamen	L	NAANAAG	2136.725342	2309.810791	382.385590	2231.440430
sub-01	ses-01	synthseg	NaN	12	Left-Putamen	L	GPCPCh	435.665131	448.210663	39.053650	444.412476
sub-01	ses-01	synthseg	NaN	12	Left-Putamen	L	CrPCr	1267.678711	1296.022949	128.737793	1293.637451
sub-01	ses-01	synthseg	NaN	12	Left-Putamen	L	GluGln	1530.674316	1528.963135	184.223709	1556.237183
sub-01	ses-01	synthseg	NaN	12	Left-Putamen	L	Ins	1010.337097	1018.918762	98.640350	1026.011719
sub-01	ses-01	synthseg	NaN	13	Left-Pallidum	L	NAANAAG	2034.484375	2078.768066	307.048096	2110.898438
sub-01	ses-01	synthseg	NaN	13	Left-Pallidum	L	GPCPCh	447.618805	453.608521	17.511766	450.032379
sub-01	ses-01	synthseg	NaN	13	Left-Pallidum	L	CrPCr	1254.361938	1273.921997	42.989773	1260.146484
sub-01	ses-01	synthseg	NaN	13	Left-Pallidum	L	GluGln	1431.598999	1473.228882	101.604294	1454.999146
sub-01	ses-01	synthseg	NaN	13	Left-Pallidum	L	Ins	982.183655	988.123108	26.691330	984.325012
sub-01	ses-01	synthseg	NaN	16	Brain-Stem	NaN	NAANAAG	1321.301880	1364.280273	586.337402	1436.737305
sub-01	ses-01	synthseg	NaN	16	Brain-Stem	NaN	GPCPCh	288.841522	312.781708	95.520370	305.960724
sub-01	ses-01	synthseg	NaN	16	Brain-Stem	NaN	CrPCr	1038.990967	1105.085571	395.701111	1111.731323
sub-01	ses-01	synthseg	NaN	16	Brain-Stem	NaN	GluGln	1367.851318	1422.908569	505.210144	1461.026367
sub-01	ses-01	synthseg	NaN	16	Brain-Stem	NaN	Ins	916.017456	967.563965	289.037476	958.300476
sub-01	ses-01	synthseg	NaN	17	Left-Hippocampus	L	NAANAAG	1376.004272	1448.197388	521.299988	1531.953247
sub-01	ses-01	synthseg	NaN	17	Left-Hippocampus	L	GPCPCh	304.784271	305.012909	68.495049	322.132599
sub-01	ses-01	synthseg	NaN	17	Left-Hippocampus	L	CrPCr	1072.939575	1193.332153	298.414612	1149.397095
sub-01	ses-01	synthseg	NaN	17	Left-Hippocampus	L	GluGln	1323.309448	1350.020630	411.947845	1437.278931
sub-01	ses-01	synthseg	NaN	17	Left-Hippocampus	L	Ins	934.981628	1010.880432	214.109146	982.094971
sub-01	ses-01	synthseg	NaN	18	Left-Amygdala	L	NAANAAG	898.944275	1046.354980	475.616516	973.096069
sub-01	ses-01	synthseg	NaN	18	Left-Amygdala	L	GPCPCh	260.297241	287.827789	85.536095	273.582123
sub-01	ses-01	synthseg	NaN	18	Left-Amygdala	L	CrPCr	719.540283	832.746033	298.823395	765.050232
sub-01	ses-01	synthseg	NaN	18	Left-Amygdala	L	GluGln	862.518555	984.120178	354.406158	917.175964
sub-01	ses-01	synthseg	NaN	18	Left-Amygdala	L	Ins	620.236145	696.140686	224.480240	654.914246

Connectivity

Connectivity matrix (spearman)



MRSinMRS

Parameter	Value	Unit
AcquisitionDurationMS	389	ms
Averages	1	
AveragingMode	N.A.	
CoilElements	N.A. (synthetic)	
CombinedMetabolites	tNAA (NAA+NAAG), tCr (Cr+PCr), Cho (GPC+PCh), Ins (mI), Glx (Glu+Gln)	
CompressedSensingAccelerationFactor	N.A. (synthetic)	
DataProcessing	Model-synthesized signal (3D U-Net trained on real template subjects); no k-space reconstruction	
DeltaFrequencyPPM	0.0	ppm
Dimension	3D	
FIDPointsVectorSize	512	
FOV	220 x 220 x 130	mm
FlipAngle	45.0	°
MagneticFieldStrength	3.0	T
MatrixSize	44 x 44 x 25	

Measurements	1	
MetaboliteBasisSetLCModel	NAA, NAAG, Cr, PCr, GPC, PCh, mI, sI, Glu, Gln, Lac, GABA, GSH, Tau, Asp, Ala	
Note	SynthMRSI-Project is a synthetic dataset: real T1w anatomicals paired with model-synthesized MRSI metabolite signal (3D U-Net; see MRSIPrep manuscript, Validation Datasets). No real MRSI acquisition underlies these values -- numeric geometry/timing fields (TE, RepetitionTime, FOV, resolution, matrix size, field strength, etc.) record the acquisition protocol the synthetic signal generation was matched to; fields describing a specific real pulse sequence, reconstruction, or hardware are marked N.A. since none was used.	
Orientation	Transverse	
PhaseEncoding	N.A. (synthetic)	
PreparationScans	N.A. (synthetic)	
QualityMetrics	SNR, CRLB, FWHM (synthesized, not derived from spectral fitting)	
Quantification	N.A. (synthetic; not derived from spectral fitting)	
RFCoils	N.A. (synthetic)	
RemoveOversampling	N.A.	
RepetitionTime	0.457	s
ResolutionMM3	5 x 5 x 5.2	mm
RotationDeg	-0.79	°
SaturationBands	N.A. (synthetic)	
Scanner	N.A. (synthetic; no scanner)	
Sequence	N.A. (synthetic; not acquired with a real MRSI sequence)	
ShimMode	N.A. (synthetic)	
SlabThicknessMM	95	mm
Slabs	1	
SpectralBandwidthHz	1320	Hz
SpectralSuppr	N.A. (synthetic)	
TE	0.78	s
WaterReferenceFIDPoints	512	
WaterReferenceFlipAngle	45.0	°
WaterReferenceResolutionMM3	10.0 x 10.0 x 10.0	mm
WaterReferenceTE	0.72	s
WaterReferenceTR	0.46	s
WaterSuppr	N.A. (synthetic)	
WaterSupprBWHz	N.A. (synthetic)	Hz

PrepParams

Processing parameters

Parameter	Value
Nucleus	1H
Tissue backend	synthseg-fast
Registration backend	ants
Registration T1w target	brain
Normalization	simple
Output spaces	MNI152NLin2009cAsym
Parcellation mode	synthseg
Atlas	chimera-LFMIHIFIS_scale3
SNR min	4.0
Linewidth max	0.1
CRLB max	20.0
Biharmonic spike filtering	True
Spike percentile	99.0
PVC disabled	False
T1 saturation correction	none
Connectivity	True

nproc x nthreads	1 x 16

Pipeline stages

`Validate inputs` → `Tissue segmentation` → `Anatomical prep` → `MRSI preprocessing` → `Registration` → `Tissue probability maps` → `Tissue QC` → `Partial volume correction` → `Resampling` → `Leakage QC` → `SynthSeg parcellation QC` → `Parcellation` → `Regional extraction` → `Connectivity` → `Metabolite profiles` → `Reports`

Runtime

Per-step duration

nproc: `1` | nthreads: `16`

Excludes this report-generation step's own duration (not yet known while it's still running) and any time spent before this recording's pipeline started (e.g. queued behind other recordings under `--nproc`).

PROC computed this run · **REUSED** existing outputs reused (re-run with `--overwrite` to recompute) · **N/A** not requested by this configuration · **FAILED** raised an error

Step	Outcome	Duration	% of total
Tissue segmentation	REUSED	0.0s	0.0%
Anatomical preparation	PROC	0.0s	0.0%
MRSI preprocessing	REUSED	3.9s	27.5%
MRSI-to-T1w registration	REUSED	0.0s	0.0%
T1w-to-template registration	REUSED	1.5s	10.7%
Tissue probability maps in MRSI space	REUSED	0.0s	0.0%
Partial volume correction	REUSED	0.0s	0.0%
Resampling MRSI maps to T1w/MNI space	REUSED	5.4s	38.5%
Signal leakage QC	PROC	0.0s	0.1%
SynthSeg parcellation and QC	REUSED	2.0s	14.4%
Regional metabolite extraction	PROC	0.6s	4.4%
Regional metabolic profiles and connectivity	PROC	0.6s	4.4%
T1 saturation correction --t1-correction none	N/A	-	-
Parcellation (chimera/atlas) parcellation_mode=synthseg, requires chimera or atlas	N/A	-	-
Total (through report generation)		**14.1s**	100.0%

Outputs

```
ses-01/
├── anat/
│   └── synthseg/
│       ├── sub-01_ses-01_atlas-synthseg_desc-parcelqc.tsv
│       ├── sub-01_ses-01_atlas-synthseg_desc-robustGMWM.tsv
│       ├── sub-01_ses-01_space-mrsi_atlas-synthseg_desc-regional_metabolites.tsv
│       ├── sub-01_ses-01_space-mrsi_atlas-synthseg_desc-robustGMWM_dseg.nii.gz
│       ├── sub-01_ses-01_space-T1w_atlas-synthseg_desc-robustGMWM_dseg.nii.gz
│       ├── sub-01_ses-01_space-T1w_desc-brainCSF_T1w.nii.gz
│       ├── sub-01_ses-01_space-T1w_desc-brainCSFmask_mask.nii.gz
│       ├── sub-01_ses-01_space-T1w_desc-synthsegBrain_mask.nii.gz
│       ├── sub-01_ses-01_space-T1w_desc-synthsegBrain_T1w.nii.gz
│       ├── sub-01_ses-01_space-T1w_desc-synthsegFastInput_T1w.nii.gz
│       └── sub-01_ses-01_space-T1w_desc-synthsegParcRobust_dseg.nii.gz
├── confounds/
│   ├── sub-01_ses-01_desc-leakageqc.tsv
│   ├── sub-01_ses-01_desc-mrsiqc.tsv
│   ├── sub-01_ses-01_desc-ventricleqc.tsv
│   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_desc-fwhm_mrsi.nii.gz
│   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_desc-snr_mrsi.nii.gz
│   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-CrPCr_desc-crlb_mrsi.nii.gz
│   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-GluGln_desc-crlb_mrsi.nii.gz
│   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-GPCPCh_desc-crlb_mrsi.nii.gz
│   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-Ins_desc-crlb_mrsi.nii.gz
│   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-NAANAAG_desc-crlb_mrsi.nii.gz
│   ├── sub-01_ses-01_space-mrsi_desc-brain_mask.nii.gz
│   ├── sub-01_ses-01_space-mrsi_label-CSF_probseg.nii.gz
│   ├── sub-01_ses-01_space-mrsi_label-GM_probseg.nii.gz
│   ├── sub-01_ses-01_space-mrsi_label-latvent_mask.nii.gz
│   ├── sub-01_ses-01_space-mrsi_label-WM_probseg.nii.gz
│   ├── sub-01_ses-01_space-mrsi_met-CrPCr_desc-qcmask_mask.nii.gz
```

```
│   ├── sub-01_ses-01_space-mrsi_met-CrPCr_desc-spikemask_mask.nii.gz
│   ├── sub-01_ses-01_space-mrsi_met-GluGln_desc-qcmask_mask.nii.gz
│   ├── sub-01_ses-01_space-mrsi_met-GluGln_desc-spikemask_mask.nii.gz
│   ├── sub-01_ses-01_space-mrsi_met-GPCPCh_desc-qcmask_mask.nii.gz
│   ├── sub-01_ses-01_space-mrsi_met-GPCPCh_desc-spikemask_mask.nii.gz
│   ├── sub-01_ses-01_space-mrsi_met-Ins_desc-qcmask_mask.nii.gz
│   ├── sub-01_ses-01_space-mrsi_met-Ins_desc-spikemask_mask.nii.gz
│   ├── sub-01_ses-01_space-mrsi_met-NAANAAG_desc-qcmask_mask.nii.gz
│   ├── sub-01_ses-01_space-mrsi_met-NAANAAG_desc-spikemask_mask.nii.gz
│   ├── sub-01_ses-01_space-orig_desc-fwhm_mrsi.nii.gz
│   ├── sub-01_ses-01_space-orig_desc-snr_mrsi.nii.gz
│   ├── sub-01_ses-01_space-orig_met-CrPCr_desc-crlb_mrsi.nii.gz
│   ├── sub-01_ses-01_space-orig_met-GluGln_desc-crlb_mrsi.nii.gz
│   ├── sub-01_ses-01_space-orig_met-GPCPCh_desc-crlb_mrsi.nii.gz
│   ├── sub-01_ses-01_space-orig_met-Ins_desc-crlb_mrsi.nii.gz
│   ├── sub-01_ses-01_space-orig_met-NAANAAG_desc-crlb_mrsi.nii.gz
│   ├── sub-01_ses-01_space-T1w_label-CSF_probseg.nii.gz
│   ├── sub-01_ses-01_space-T1w_label-GM_probseg.nii.gz
│   └── sub-01_ses-01_space-T1w_label-WM_probseg.nii.gz
├── connectivity/
│   ├── sub-01_ses-01_atlas-synthseg_npert-50_filt-biharmonic_pvcorr_GM_desc-connectivity_mrsi.npz
│   ├── sub-01_ses-01_atlas-synthseg_npert-50_filt-biharmonic_pvcorr_GM_desc-connectivity_mrsi.tsv
│   ├── sub-01_ses-01_atlas-synthseg_npert-50_filt-biharmonic_pvcorr_GM_desc-edges_mrsi.tsv
│   ├── sub-01_ses-01_atlas-synthseg_npert-50_filt-biharmonic_pvcorr_GM_desc-metabolicprofiles_mrsi.npz
│   └── sub-01_ses-01_atlas-synthseg_npert-50_filt-biharmonic_pvcorr_GM_desc-nodes_mrsi.tsv
├── logs/
│   └── sub-01_ses-01_desc-mrsiprep_log.txt
├── mrsi/
│   ├── mni/
│   │   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-CrPCr_desc-signal_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-GluGln_desc-signal_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-GPCPCh_desc-signal_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-Ins_desc-signal_mrsi.nii.gz
│   │   └── sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-NAANAAG_desc-signal_mrsi.nii.gz
│   ├── orig/
│   │   ├── sub-01_ses-01_space-mrsi_desc-reference_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-CrPCr_desc-signal_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-CrPCr_desc-signalspikefilt_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-GluGln_desc-signal_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-GluGln_desc-signalspikefilt_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-GPCPCh_desc-signal_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-GPCPCh_desc-signalspikefilt_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-Ins_desc-signal_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-Ins_desc-signalspikefilt_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-NAANAAG_desc-signal_mrsi.nii.gz
│   │   └── sub-01_ses-01_space-mrsi_met-NAANAAG_desc-signalspikefilt_mrsi.nii.gz
│   ├── orig-pvc/
│   │   ├── sub-01_ses-01_space-mrsi_met-CrPCr_desc-signalpvc_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-GluGln_desc-signalpvc_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-GPCPCh_desc-signalpvc_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-Ins_desc-signalpvc_mrsi.nii.gz
│   │   └── sub-01_ses-01_space-mrsi_met-NAANAAG_desc-signalpvc_mrsi.nii.gz
│   └── parcel/
│       └── sub-01_ses-01_atlas-synthseg_desc-GMmetprofiles_mrsi.npz
└── transforms/
    ├── anat/
    │   ├── sub-01_ses-01_desc-t1w_to_mni.affine.mat
    │   ├── sub-01_ses-01_desc-t1w_to_mni.affine_inv.mat
    │   ├── sub-01_ses-01_desc-t1w_to_mni.syn.nii.gz
    │   └── sub-01_ses-01_desc-t1w_to_mni.syn_inv.nii.gz
    └── mrsi/
        ├── sub-01_ses-01_desc-mrsi_to_t1w.affine.mat
        ├── sub-01_ses-01_desc-mrsi_to_t1w.affine_inv.mat
        ├── sub-01_ses-01_desc-mrsi_to_t1w.syn.nii.gz
        └── sub-01_ses-01_desc-mrsi_to_t1w.syn_inv.nii.gz
```

Citations

If you use MRSIPrep, please cite: Lucchetti, F., Céléreau, E., Jenni, R., Ledoux, J.-B., Eliez, S., Delavari, F., Hagmann, P., Alemán-Gómez, Y., Klauser, A., & Klauser, P. (2026). MRSIPrep: A Standardized Post-Quantification Framework for Preprocessing Whole-Brain Magnetic Resonance Spectroscopic Imaging. *bioRxiv*. doi:10.64898/2026.09.11.750896. A downloadable BibTeX entry is in `CITATION.cff` in the MRSIPrep repository.

MRSI acquisition reporting follows the MRSinMRS minimum reporting standard (Lin et al. 2021).